

Brain Tumor MRI Classification Model

Field Guide for Clinical and Operational Use

Model: ResNet18 pretrained image classifier

Use case: MRI slice classification into glioma, meningioma, or pituitary tumor

Final Project – MPHY 6120 AI in Medicine

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Status: Supporting image review, not for standalone diagnosis

Intended use statement

This tool is designed to support image review by providing a preliminary tumor class prediction and visual explanation. It should not be used as the only basis for diagnosis, treatment planning, or patient management decisions.

1. Executive Summary

This tool is a deep learning based image classification model developed to support the review of brain tumor MRI images. The model takes one MRI brain slice as input and predicts whether the image is most consistent with glioma, meningioma, or pituitary tumor. In addition to the predicted label, the tool provides a confidence score and visual explanation outputs, including Grad-CAM, Grad-CAM++, and occlusion sensitivity maps. These outputs are included to help the clinical user understand where the model is looking, not to replace clinical interpretation.

The primary users are radiologists, radiation oncologists, and clinical AI implementation teams. In a clinical workflow, the model should be used as a second reader or triage support tool after MRI acquisition and before final clinical sign-off. It may also be helpful for education, quality review, and early research evaluation where users want to compare model attention with visible tumor features.

The internal test results show strong model performance. In the held-out test set of 909 images, the model correctly classified 902 images, corresponding to an overall accuracy of 99% with an approximate 95% confidence interval of 98.4% to 99.6%. The ROC-AUC was approximately 1.00 for all three classes, and average precision was approximately 0.999 to 1.000. The calibration analysis showed a low Brier score of 0.0035, suggesting strong overall probability performance, although the reliability curve still shows that some probability bins are unstable and the model can be overconfident.

Overall, this model is promising as a clinical decision support tool. However, the current evidence is based on internal retrospective image data and formal human-in-the-loop review is needed. Final clinical responsibility must remain with the qualified clinician.

Category	Result	Operational meaning
Input	Single 2D MRI brain slice	The model does not evaluate full 3D exams or clinical history.
Output	Glioma, meningioma, or pituitary tumor plus confidence score	Use as decision support only.
Internal test set	909 images	Held-out dataset used for final evaluation.
Overall accuracy	99% (902/909), 95% CI 98.4%-99.6%	Very high internal performance.
Discrimination	ROC-AUC approximately 1.00 for all classes	The model separates the three classes very well on this dataset.
Calibration	Brier loss 0.0038	Predicted probabilities are generally strong, but some bins are unstable.

2. Clinical Context

Clinical problem

Brain tumor classification is a common and important task in neuro-oncology imaging. In routine practice, brain MRI is interpreted by radiologists who assess tumor location, enhancement pattern, edema, mass effect, relationship to surrounding structures, and other imaging features. The interpretation then contributes to clinical decisions such as referral pathway, surgical planning, biopsy planning, radiation therapy evaluation, and patient counseling. However, tumor classification can still be challenging when image quality is limited, when only a small number of slices show the lesion clearly, or when tumors have overlapping imaging appearances.

This model addresses a focused part of that larger clinical problem. It does not attempt to replace the full radiology interpretation. Instead, it provides a fast prediction for three broad tumor classes: glioma, meningioma, and pituitary tumor. The practical value is not only the predicted class, but also the ability to provide a consistent second read and a visual explanation that can be reviewed by a human user.

Current standard of care

The current standard of care is expert interpretation of MRI by trained clinicians, usually radiologists, supported by clinical history and prior imaging when available. For many patients, the final diagnosis may require pathology from biopsy or resection. Imaging classification is therefore an important clinical input, but it is not always the final diagnostic truth.

In the current workflow, the radiologist reviews the complete MRI examination, including multiple sequences and planes. A final report is generated based on imaging findings, comparison studies, and clinical context. Any AI model used in this setting must respect this workflow. It should provide additional information without hiding uncertainty or forcing the clinician to accept the model prediction.

Current workflow step	Current practice	AI-supported change
MRI acquisition	Images acquired using local scanner protocol.	No change. Image quality remains the responsibility of the imaging team.
Image review	Radiologist reviews full exam and prior imaging.	Model provides a preliminary class prediction for selected slices.
Interpretation	Clinician integrates imaging features and clinical context.	Clinician compares model prediction and heatmap with visible anatomy.
Reporting	Final report signed by clinician.	AI result may support review, but should not automatically populate final diagnosis.
Quality review	Errors reviewed if clinically detected.	Model performance can be monitored continuously with predefined thresholds.

Expected workflow impact

The most reasonable workflow role for this model is as a decision support layer. It may help prioritize cases, provide a second opinion for common tumor classes, and encourage consistent review of imaging features. For example, when the model predicts glioma with high confidence and the heatmap overlaps the suspicious lesion, the tool may increase reader confidence. When the model prediction disagrees with the radiologist impression or the heatmap focuses outside the lesion, the case should be reviewed more carefully rather than automatically accepted.

Clinical workflow principle

The model should be treated like an assistant that flags a possibility. The radiologist or responsible clinician remains the final decision maker.

3. Technical Description

Data requirements

The model was developed using MRI brain tumor images stored as 2D image files. The dataset used in this project contained 4,239 training images, 908 validation images, and 909 test images. The three target labels were glioma, meningioma, and pituitary tumor. The training split was balanced, with each class representing approximately one third of the training images. This balanced class distribution is important because it reduces the risk that the model simply learns to favor the most common class.

Dataset component	Count / description
Training images	4,239
Validation images	908
Test images	909
Classes	Glioma, meningioma, pituitary tumor
Estimated unique numeric IDs	2,004
Input format used for modeling	Single MRI slice image, resized and normalized for ResNet18 input

Before training, the images were resized and normalized. Data augmentation was applied during training. The augmentation examples included flip, rotation, and intensity jitter. The purpose of augmentation is to reduce overfitting, teach the model to ignore small acquisition differences that are not clinically meaningful, and improve generalization compared with training longer on a small fixed dataset.

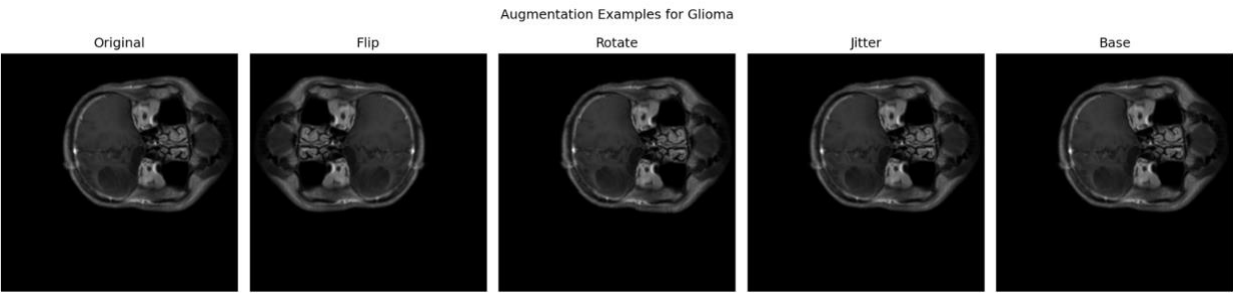


Figure 1. Example MRI augmentation used during training.

Model architecture

In this project, transfer learning was used. The model is based on a pretrained ResNet18 convolutional neural network. ResNet18 is a widely used image classification architecture that learns image features through a series of convolutional layers. This means that the model starts from weights learned from a large general image dataset, then the final classifier is adapted to the brain tumor classification task. This approach is useful when the medical dataset is smaller than the very large datasets usually needed to train deep learning models from scratch.

At a high level, the early layers detect simple features such as edges and intensity transitions, while deeper layers combine these features into more task-specific image patterns. The first convolutional layer filters show that the model learned low-level image patterns such as edges, gradients, and texture-like structures. These features are expected for an image classification model, but they do not by themselves prove that the model understands tumor biology.

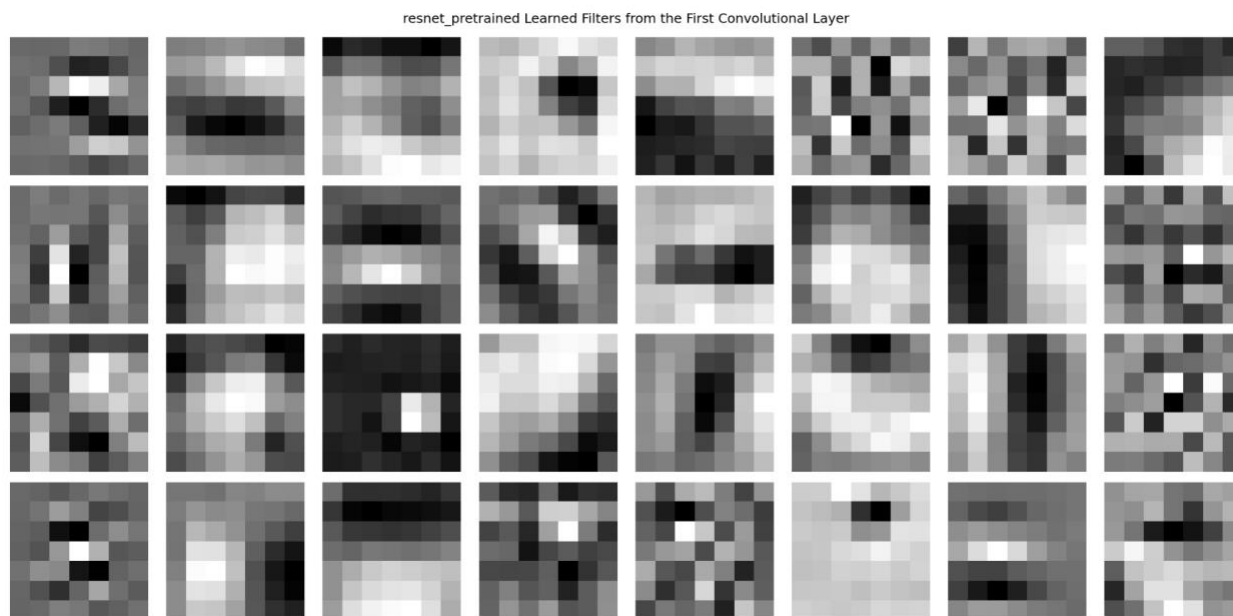


Figure 2. Learned filters from the first convolutional layer of the pretrained ResNet18 model.

Input and output specification

Item	Specification
Input image	One .jpg MRI brain slice image, preprocessed to the model input size and channel format.
Input assumptions	Image should be representative of the tumor region and similar in appearance to the training data.
Primary output	Predicted tumor class: glioma, meningioma, or pituitary tumor.
Secondary output	Confidence score for the predicted class.
Explainability outputs	Grad-CAM, Grad-CAM++, and occlusion sensitivity maps.
Unsupported inputs	Full 3D MRI exams, multi-sequence studies, DICOM metadata, non-brain images, post-operative images not represented in training, or classes outside the three target labels.

Computational requirements

Training is best performed on a GPU because model optimization requires repeated forward and backward passes through the network. Inference can usually be performed on a CPU or GPU because only one forward pass is required per image. For operational deployment, the main requirements are stable preprocessing, version-controlled model weights, input validation, logging of predictions, and a secure environment that complies with local institutional requirements.

Component	Recommended requirement
Training hardware	GPU recommended.
Inference hardware	CPU acceptable for low-volume use; GPU preferred for high-throughput workflows.
Software stack	Python, PyTorch, image preprocessing pipeline, model weight file, logging module.
Operational controls	Model versioning, data provenance, prediction logs, and periodic performance review.

4. Performance Evaluation

Evaluation dataset and endpoint

The model was evaluated on a held-out internal test set of 909 images. The endpoint was correct classification into one of three tumor classes. Because the evaluation was performed on image slices rather than full patient exams, the results should be interpreted as image-level performance. Patient-level performance may be different and should be tested separately before clinical deployment.

Primary performance metrics

Metric	Result	Comment
Overall accuracy	99% (902/909), 95% CI 98.4%-99.6%	Strong internal test performance.
Glioma recall	100.0% (301/301), 95% CI 98.7%-100.0%	No glioma cases were missed in this test set.
Meningioma recall	98.7% (296/300), 95% CI 96.6%-99.5%	Four meningioma images were misclassified.
Pituitary tumor recall	99.0% (305/308), 95% CI 97.2%-99.7%	Three pituitary images were misclassified.
ROC-AUC	Approximately 1.00 for all classes	Confidence intervals were not calculated for AUC in this run.
Average precision	Approximately 0.999-1.000	Precision-recall curves show very strong separation.

Metric	Result	Comment
Brier score	0.0035	Low value, but calibration bins still show instability.

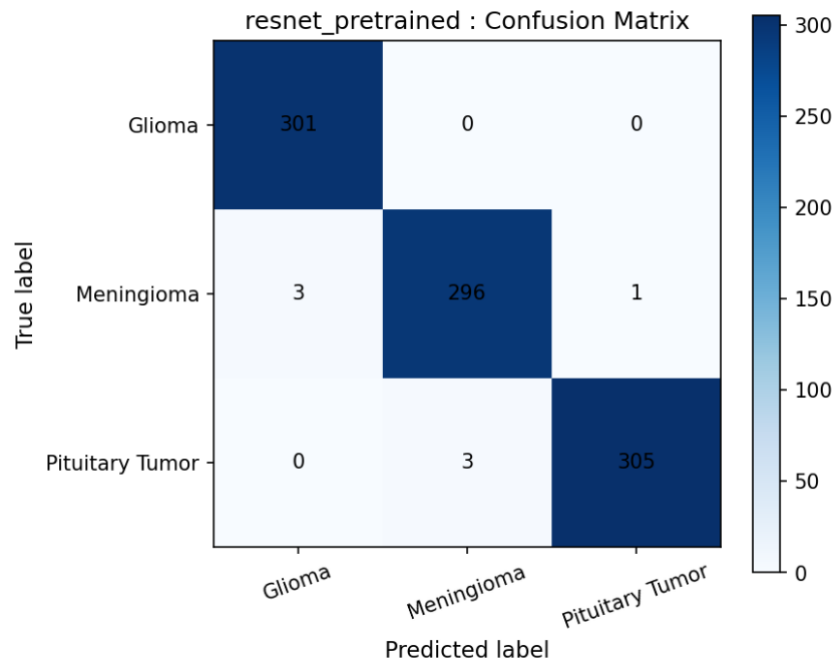


Figure 3. Confusion matrix for the ResNet18 pretrained model on the internal test set.

Class-level findings

The model performed best for glioma in this evaluation, with all glioma images correctly classified. The main remaining errors were between meningioma and glioma, and between pituitary tumor and meningioma. These errors are clinically plausible because some tumor appearances can overlap on single slices, especially when the model does not have access to full multi-sequence MRI context or clinical history.

True class	Correct	Most important observed error	Operational interpretation
Glioma	301/301	None observed in the test set	Strong internal performance, but should be externally validated.
Meningioma	296/300	Three predicted as glioma and one predicted as pituitary tumor	Review meningioma cases carefully when confidence is moderate or heatmap is off-target.
Pituitary tumor	305/308	Three predicted as meningioma	Pituitary region anatomy and limited field of view may affect interpretation.

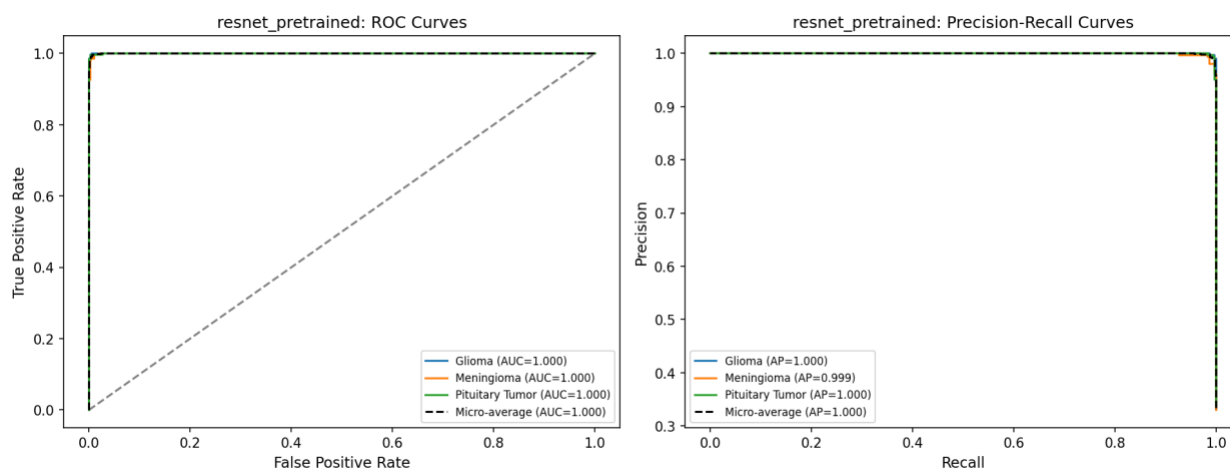


Figure 4. ROC and precision-recall curves for the ResNet18 pretrained model.

Calibration and confidence behavior

The model confidence is high for many examples. The Brier score is low, which suggests good overall probability performance. However, the calibration curve should be interpreted carefully. Some mid-range probability bins show noisy behavior because relatively few examples fall in those bins. This means the model may still appear overconfident or underconfident in specific confidence ranges. In a clinical implementation, confidence should therefore be used as a support signal, not as a direct measure of diagnostic certainty.

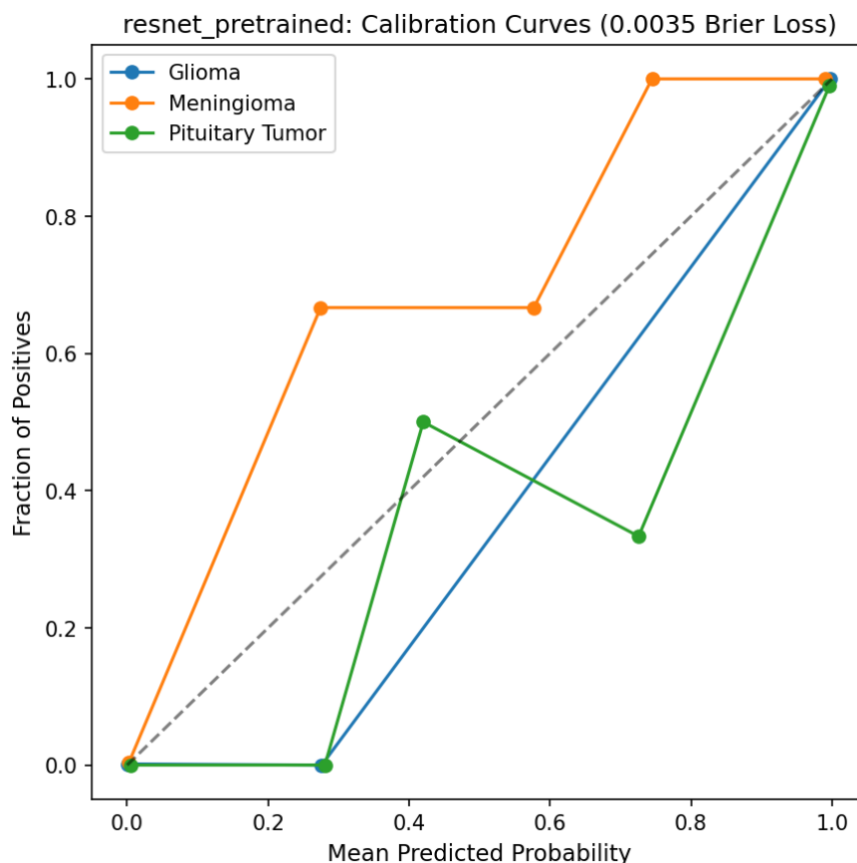


Figure 5. Calibration curve and Brier score for the ResNet18 pretrained model

Explainability review

The explainability review used Grad-CAM, Grad-CAM++, and occlusion sensitivity. These methods provide different views of model attention. Grad-CAM highlights areas associated with the predicted class based on internal feature maps. Occlusion sensitivity tests how much the prediction changes when different regions of the image are covered. In the reviewed examples, the occlusion maps often highlighted focal image regions that appear clinically relevant. However, some maps also showed attention outside the tumor area or across broad background regions.

This is important because a correct prediction does not automatically mean the model used the correct image features. For clinical review, a prediction should be considered more trustworthy when the confidence is reasonable and the heatmap overlaps the lesion or clinically meaningful anatomy. If the heatmap is mostly outside the lesion, the result should be treated with caution even if the predicted label is correct.

resnet_pretrained: Occlusion Sensitivity

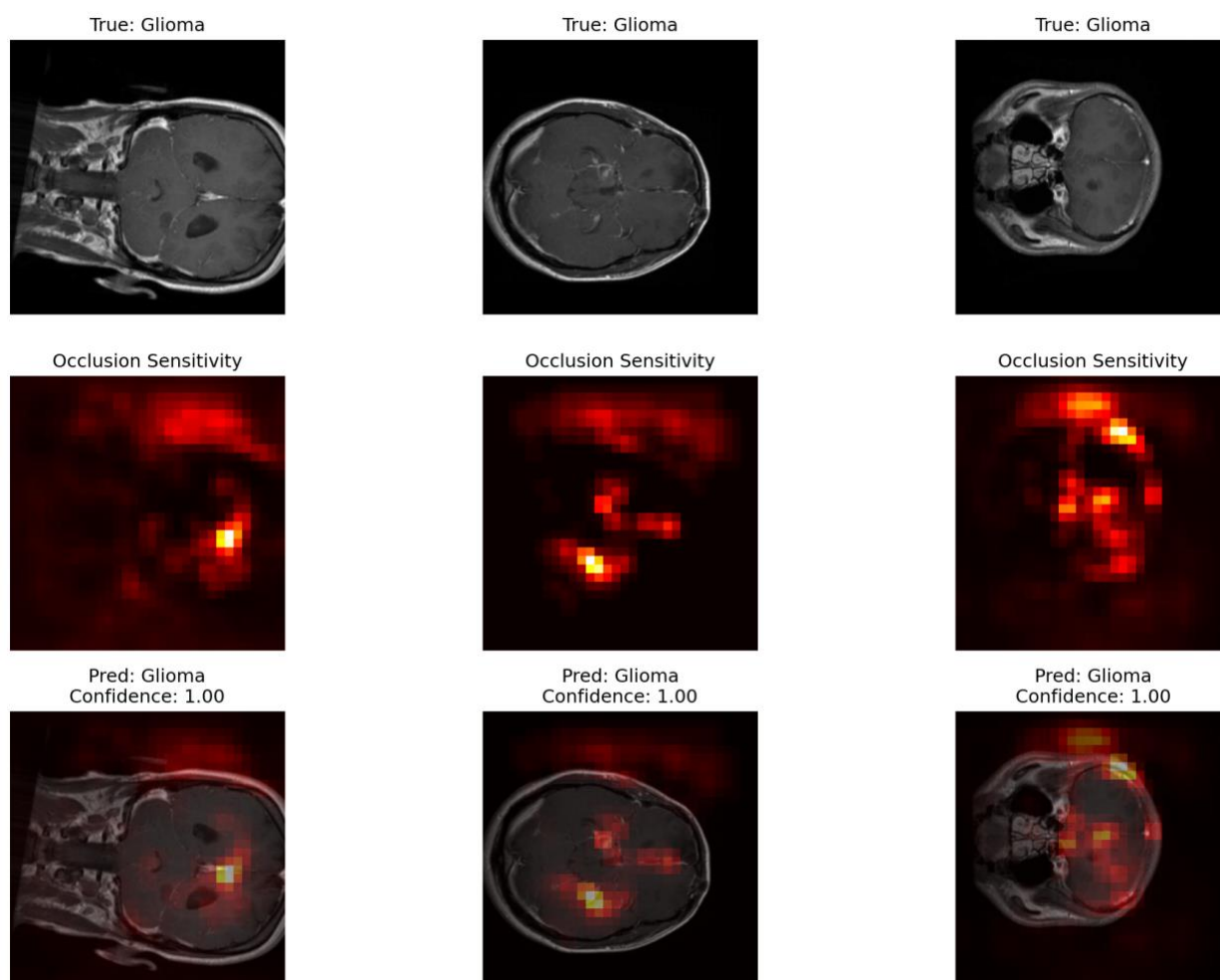


Figure 6. Occlusion sensitivity examples showing regions that influence the model prediction.

Early Layer Grad-CAM vs Grad-CAM++ (Full Run)

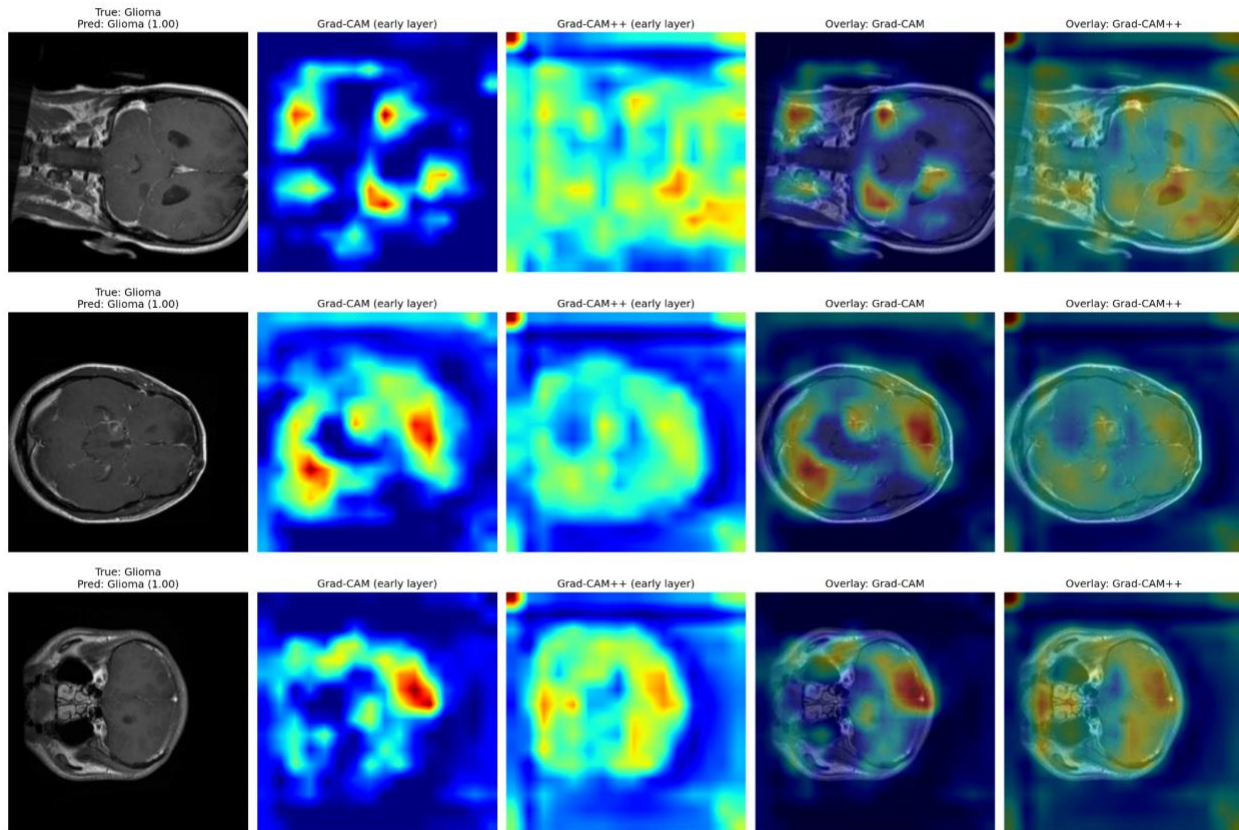


Figure 7. Early-layer Grad-CAM and Grad-CAM++ comparison examples.

Known failure modes

Failure mode	Observed evidence	Mitigation
Meningioma classified as glioma	Three meningioma images were predicted as glioma.	Flag moderate-confidence cases for additional human review.
Pituitary tumor classified as meningioma	Three pituitary images were predicted as meningioma.	Review sellar and parasellar anatomy carefully.
Overconfidence	Many correct and incorrect examples have high confidence; calibration curve has unstable bins.	Use confidence thresholds and escalation rules.
Heatmap outside lesion	Grad-CAM and occlusion maps may highlight broad or non-tumor regions.	Require human review of explanation maps before clinical trust.
No segmentation mask validation	No matched masks were available, so IoU and Dice were not calculated.	Collect lesion masks for future explanation validation.

5. Governance Framework

Acceptance testing protocol for new sites

Before this tool is used at any new site, the site should complete a formal acceptance test using local MRI images. The purpose is to confirm that the model behaves safely with the local scanner protocols, image preprocessing pathway, and patient population. The acceptance test should be completed before the model is connected to routine clinical workflow.

1. Define the intended use at the site, including who can view the model output and where it appears in the workflow.
2. Create a locked test dataset from local MRI images with confirmed labels. The test dataset should include all three classes and should not be used for model training.
3. Verify preprocessing, including image orientation, resizing, channel conversion, intensity normalization, and file handling.
4. Run the locked model version on the local test dataset without changing model weights.
5. Compare model predictions against local reference labels and calculate accuracy, recall, precision, and confusion matrix.
6. Review a representative sample of explanation maps with clinical users.
7. Document pass or fail status, risks, required mitigations, and approval signatures before go-live.

Acceptance item	Suggested minimum criterion	Action if not met
Overall accuracy	At least 90% on local acceptance dataset	Investigate data mismatch and pause go-live.
Per-class recall	No class below 85% without documented clinical justification	Increase test set size, inspect errors, and consider retraining or restricting use.
Image preprocessing	100% successful preprocessing with no orientation or scaling defects	Fix pipeline before further testing.
Explanation review	Most reviewed heatmaps should overlap lesion or clinically relevant anatomy	Restrict use and collect more validation data.
User training	All users trained on intended use and limitations	Do not release into clinical workflow.

Monitoring plan

After deployment, monitoring should focus on model performance, data quality, confidence behavior, and workflow impact. Monitoring is needed because clinical imaging data can drift over time. Drift may occur because of scanner upgrades, protocol changes, new patient populations, changes in image export settings, or changes in tumor case mix.

Signal to monitor	Threshold / trigger	Frequency	Owner
Prediction volume by class	Unexpected class shift greater than 20% from baseline	Monthly	AI operations lead
Low-confidence rate	Increase greater than 5 percentage points from baseline	Monthly	Clinical AI lead
Disagreement with final clinical label	Any cluster of repeated disagreement or monthly accuracy below accepted threshold	Monthly or quarterly	Radiology lead
Preprocessing failures	Any repeated failure or orientation defect	Continuous logging	Engineering support
Heatmap quality	Repeated off-target heatmaps in reviewed sample	Quarterly	Clinical reviewer and physicist
Model version and data logs	Any unapproved model or pipeline change	Continuous	Governance owner

Escalation protocol

The model should have predefined escalation rules. This is important because users should not have to decide from scratch whether a performance issue is serious enough to pause use. If the model behaves unexpectedly, the safest approach is to restrict use while the issue is investigated.

Event	Severity	Immediate action	Escalation contact
Prediction appears inconsistent with obvious image finding	Moderate	Clinician overrides model and logs case for review.	Clinical AI lead
Heatmap repeatedly focuses outside brain or outside lesion	Moderate	Continue only with caution and review additional cases.	Medical physicist / AI reviewer
Sudden performance drop or repeated class-specific errors	High	Pause routine use until root cause analysis is completed.	Radiology lead and AI operations lead
Preprocessing defect affecting image orientation or scaling	High	Pause model output immediately.	Engineering support and governance owner
Unapproved model version used clinically	High	Remove model from workflow and perform audit.	Governance owner

Human-in-the-loop requirements

Human review is required for all clinical use. The model output should be visible to trained users only, and the interface should clearly label the output as AI decision support. The user must be able to override the model without additional burden. The final report or clinical decision should not be generated automatically from the model prediction.

- A qualified clinician must confirm or reject the model prediction.
- The confidence score should be reviewed together with the image and heatmap, not in isolation.
- Discordant cases should be documented and used for periodic quality review.
- The model should not be used when the input image is outside the approved scope.
- Users should receive training on limitations, failure modes, and escalation procedures before access is granted.

6. Limitations and Risks

Known limitations

The most important limitation is that this model evaluates single 2D MRI slices rather than the full 3D MRI examination. In clinical practice, tumor classification usually depends on multiple sequences, multiple planes, prior imaging, patient history, and sometimes pathology. A single slice cannot capture the full disease context. Therefore, the model should not be interpreted as a complete diagnostic system.

- The model was evaluated on internal retrospective data only.
- External validation on new institutions, scanners, protocols, and patient populations has not yet been completed.
- The tool only predicts three classes and does not identify other brain lesions or tumor subtypes.
- The explanation maps were not quantitatively validated against segmentation masks because no matched masks were available.
- Confidence scores may appear very high even when the model is wrong or when the heatmap is not clinically convincing.

Populations and situations where performance may degrade

Performance may degrade when the input data does not look like the training data. This includes different scanner vendors, different pulse sequences, major differences in image contrast, artifacts, post-operative changes, hemorrhage, motion, pediatric cases if not represented in training, rare tumor presentations, and lesions outside the three trained categories. Local validation is especially important before using the model on a new population.

Potential for misuse

The main misuse risk is over-reliance on the model because the performance metrics appear very high. High internal accuracy does not guarantee safe performance in a new clinical setting. Another risk is treating the confidence score as diagnostic certainty. The confidence score only reflects the model output for the given image and pipeline. It does not account for clinical history, pathology, image acquisition limitations, or disease categories outside the training labels.

What this tool should not be used for

Do not use this model for these purposes

This tool should not be used as the sole basis for diagnosis, pathology replacement, treatment planning, target delineation, dose prescription, surgical decision making, or automatic report generation. It should also not be used on unsupported tumor types or low-quality images without human review.

7. Appendix

Appendix A. Detailed results table

Class	Recall	Precision	F1 score	Error summary
Glioma	100.0%	99.0%	99.5%	No false negatives; three non-glioma images predicted as glioma.
Meningioma	98.7%	99.0%	98.8%	Three false negatives to glioma and one to pituitary tumor; three pituitary images predicted as meningioma.
Pituitary tumor	99.0%	99.7%	99.3%	Three false negatives to meningioma; one meningioma image predicted as pituitary tumor.

Appendix B. Error examples

The error analysis shows examples of correct classifications and misclassifications. The misclassified examples are useful for understanding where the model should be reviewed more carefully. Several misclassified cases still had moderate to high confidence, which supports the need for human review and monitoring rather than automated use.

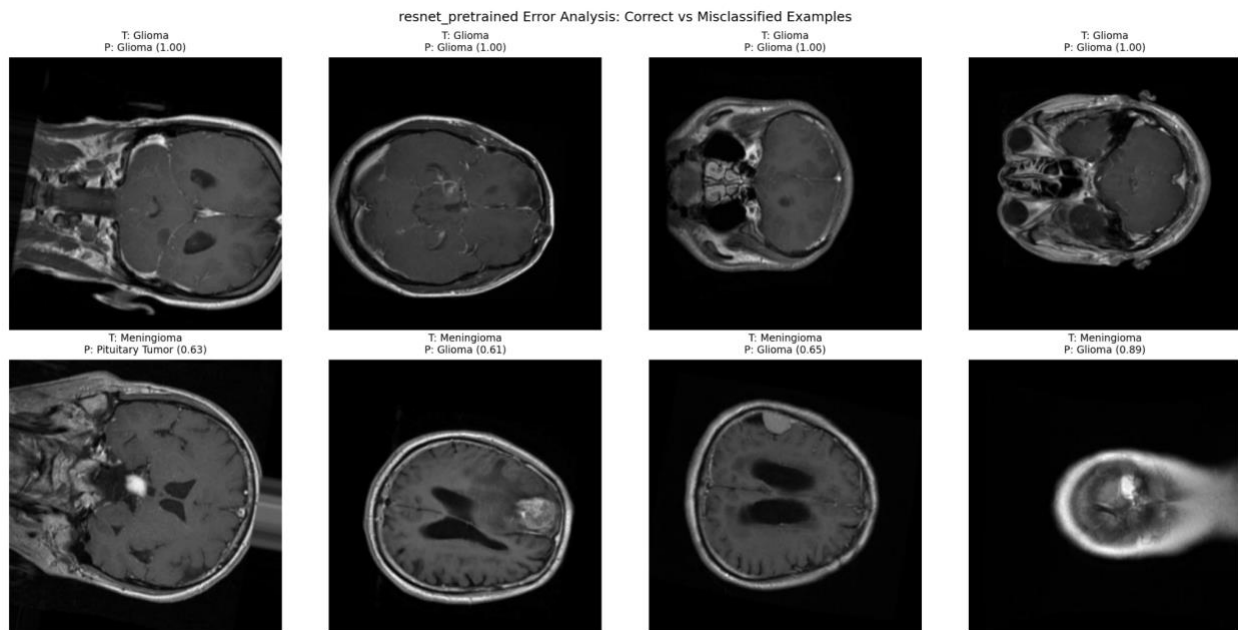


Figure 8. Correct and misclassified examples from the internal test evaluation.

Appendix C. Data review examples

The sample image grid and class distribution were used as part of the data review process. The class distribution shows a balanced training dataset. The image grid also demonstrates that images vary in

orientation, tumor size, tumor appearance, and field of view. This variation is useful for model learning, but it also means that consistent preprocessing and careful validation are needed.

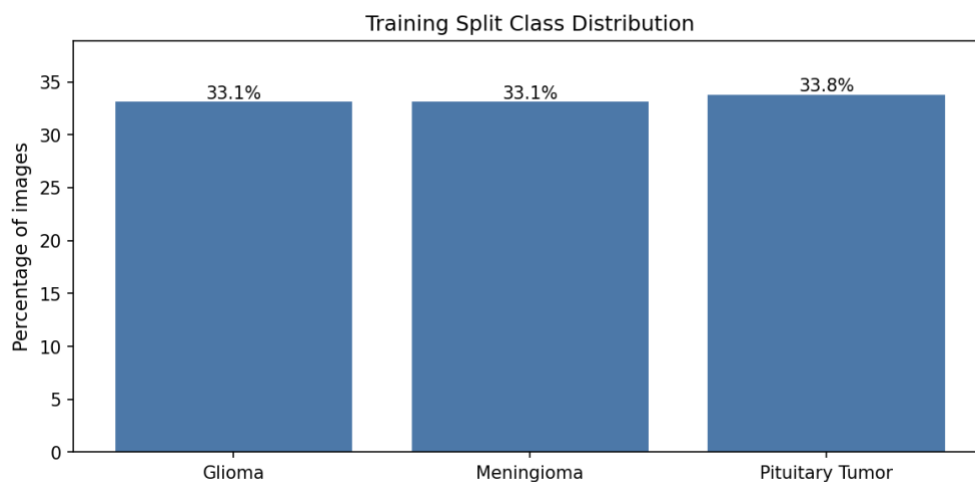


Figure 9. Training split class distribution.

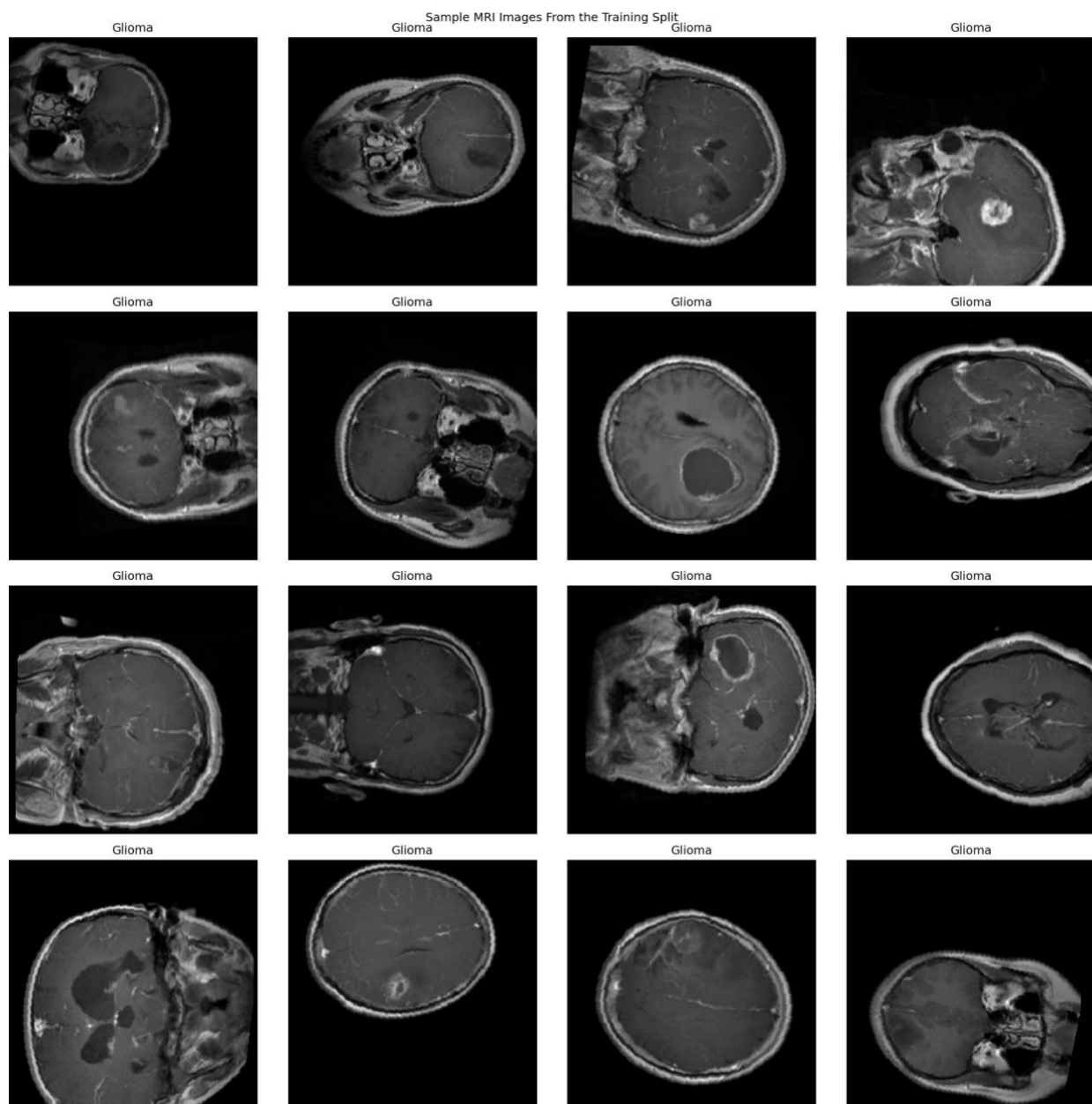


Figure 10. Sample MRI images from the training split.

Appendix D. Technical implementation notes

Area	Implementation note
Model version control	Store model weights, code commit, preprocessing settings, and class mapping together.
Class mapping	Lock the mapping between numeric output and class name to prevent reporting errors.
Preprocessing	Preprocessing must be identical between training, validation, test, and deployment.
Logging	Log image ID, model version, prediction, confidence, preprocessing

Area	Implementation note
	status, and reviewer feedback.
Security	Use institution-approved storage and access control for patient images and prediction logs.
Future improvement	Add external validation, patient-level evaluation, 3D or multi-sequence input, and lesion mask comparison for heatmaps.

Appendix E. Source artifacts used for this guide

- Data audit report: train, validation, test counts and class names.
- Training split class distribution and sample image grid.
- Augmentation example and augmentation note.
- ResNet18 pretrained training curves.
- Confusion matrix, ROC/PR curves, calibration curve, and error analysis.
- First-layer learned filters, Grad-CAM comparison, occlusion sensitivity, and mask overlap report.